

Performance Analysis Using Air Gap Defected Through Silicon Via: Impact on Crosstalk and Power

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Abstract—In the recent past, various fabrication-related defects such as electromigration-induced open/short faults, interfacial cracks, and thermal stress-induced leakage problems in through silicon via (TSV) have a major impact on the reliability and performance of a TSV-based 3-D integrated circuit (IC). Considering these facts, this article for the first time provides the equivalent resistance-inductance-capacitance (RLGC) modeling and performance analysis of air gap-defected TSVs in comparison with defect free vertical interconnect at 7 nm technology in order to address reliability concerns. Considering different manufacturing imperfections such as air cracks and void holes, the analytical equations are derived using defective parameters to analyze the feasibility and reliability of defected TSVs. Using a driver-via-load (DVL) setup, an equivalent T-type electrical modeling of cylindrical TSV is proposed considering the impact of different forms of air-cracked defects. Therefore, the propagation delay, power dissipation, power delay product, peak noise, and dynamic crosstalk are analyzed using a carbon nanotube field effect transistor (CNTFET) driver. Encouragingly, considering a partial air-cracked TSV, the peak noise is improved by 4.83% approaching toward defect-free condition.

Index Terms—Coefficient of thermal expansion (CTE), crosstalk, driver-via-load (DVL), partial air crack, spherical air void, through silicon via (TSV), uniform air crack.

I. INTRODUCTION

ONE of the key enabling technologies required to exploit the benefits of significantly miniaturized 3-D integration is the advent of through silicon via (TSV) as vertical interconnect access (VIA) [1], [2]. Apart from the benefits offered by vertical interconnects, their unstable and immature fabrication process eventually leads to various manufacturing defects in TSVs [3]. Some of the major issues need to be considered that include interfacial delamination and VIA to silicon (Si) substrate leakage due to the coefficient of thermal expansion (CTE) mismatch and thermal stress effect [4]. Although, the creation of voids and cracks in TSVs and bumps are caused by imprecise filling of the etched vias and nonuniform bump sizes [4]. Also, the formation of pinholes

occurs in the insulating layer due to improper deposition of liner [4]. Each of these defects degrades the TSV's electrical performance as well as its thermal and mechanical reliability. However, their occurrence causes the propagation of TSV signal to be attenuated or even drastically altered [5]. However, the electrical modeling and simulation of copper (Cu) based TSVs with silicon dioxide (SiO₂) have already become popular for evaluating the propagation delay, power dissipation, and crosstalk performance [6], [7], [8], [9]. Based on the Cu-based TSV fabrication outcomes, the existing research works [10], [11] reported that a realistic VIA structure incorporates SiO₂ liner that leads to potential mechanical reliability concerns that are already the challenging problem in 2-D integrated circuits (ICs) and are considerably more problematic in 3-D integration. Moreover, in current research scenario, it is desirable to manufacture and design the defect-free TSVs in order to achieve improved performance in all aspects. However, the expected level of chip yield with increasing TSV count per chip is compromised due to the presence of various defects in TSVs. Earlier, several other researchers [12], [13], [14], [15] reported on the analysis of various types of faults and examined the electrical behavior of the TSV in terms of improved test procedures and the reliability of TSV-based 3-D ICs. Gerakis et al. [12] presented an analytical electrical circuit for air-cracked (open fault) TSVs in order to develop accurate and reliable fault models to support test procedures. However, the proposed model was confined to analyzing the uniform air crack type of fault and was unable to examine other open fault scenarios. Furthermore, the same research group in [13] accurately accounted for the lumped electrical models of the defected TSVs, while considering the modeling of partially cracked and void hole defects in TSV filler. However, since the study was limited to primarily extracting the parasitic using analytical expressions, the need for evaluating test outcomes from TSV structures based on delay and power measurements was disregarded. Afterward, Jung et al. [14] extended the research to assess the impact of open defects between TSV and metal VIA, bump and upper die, short defect between fine-pitch bumps, and void formation in TSV. However, the authors limited their discussion only to the S-parameter analysis and the assessment of VIA performance in terms of equivalent latency and crosstalk were still need to be explored. In recent, Shang et al. [15] presented the electrical modeling of air void and pinhole defects, as well as analytical formulae based on fault factors (physical size and material parameters).

Manuscript received 4 November 2022; accepted 21 November 2022. Date of publication 28 November 2022; date of current version 27 December 2022. Recommended for publication by Associate Editor P. Fay upon evaluation of reviewers' comments. (Corresponding author: Maya Chandrakar.)

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Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TCPMT.2022.3225142>.

Digital Object Identifier 10.1109/TCPMT.2022.3225142

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However, the proposed method was unable to perform better in diagnosing weak manufacturing defects in TSVs. Although, the fabrication has been evolved for different shapes of TSVs but the manufacturing of complete defect-free TSVs is still a challenging task. Considering the state-of-the-art research, it is noticeable that the most existing test methods can only detect one or two types of TSV failures. The numerous varieties of defects associated with VIA have a significant impact on the reliability and feasibility of TSVs. Therefore, there is still a need for the in depth analysis of VIA electrical performance in terms of latency, peak noise, and power consumption.

To address the aforementioned research problems, this article for the first time presents an equivalent electrical model of TSVs with different types of air gap defects to analyze the VIA performance. Additionally, the model considers the shape, size, and location of a particular defect scenario. A carbon nanotube field effect transistor (CNTFET) driver is used to drive the VIA due to its good control over the gate to limit the leakage path at the lower technology [7]. Additionally, a CNTFET driver overcomes the problems of short channel effect, gate oxide thickness, surface scattering, and Joule heating that limits the efficacy of current sub-10 nm devices. Therefore, the equivalent circuits are modeled at 7 nm technology employing a coupled driver-via-load (DVL) setup with a CNTFET driver instead of a CMOS or resistive driver for an accurate analysis. In addition, the electrical circuit models are developed using a T-type network, and hence the capacitive charging current flows over half of the TSV line resulting in reduced crosstalk noise between the coupled VIA lines.

This article is organized into five different sections wherein Section I introduces the current research scenario and briefs about the work carried out. Section II describes the physical configurations and detailed analytical resistance-inductance-conductance-capacitance (RLGC) parasitic expressions for modeling of equivalent electrical models of different VIA structures. However, Section III demonstrates and compares the VIA performances of different defected TSV cases with defect-free TSV considering a DVL simulation setup. Finally, Section IV summarizes and concludes the article with a brief summary.

II. TSV ARCHITECTURE AND CIRCUIT MODEL

This section includes the physical configuration and electrical modeling of a defect free and a defected TSV with uniform air crack [16], partial air crack with a cylindrical connector [17], and spherical air void [17] at 7 nm technology. Consequently, the electrical models based on a distributed T-network are developed considering analytical RLGC expressions using VIA dimensions and material properties.

A. TSV Configuration and Equivalent RLGC Model

This section presents an accurate physical configuration and equivalent RLGC model based on the different VIA geometries. The physical architectures with electrical models of defect free, uniform air crack, partial air crack with a cylindrical connector, and spherical air void TSVs are shown in Fig. 1(a). Also, Fig. 1(b) represents a DVL setup wherein

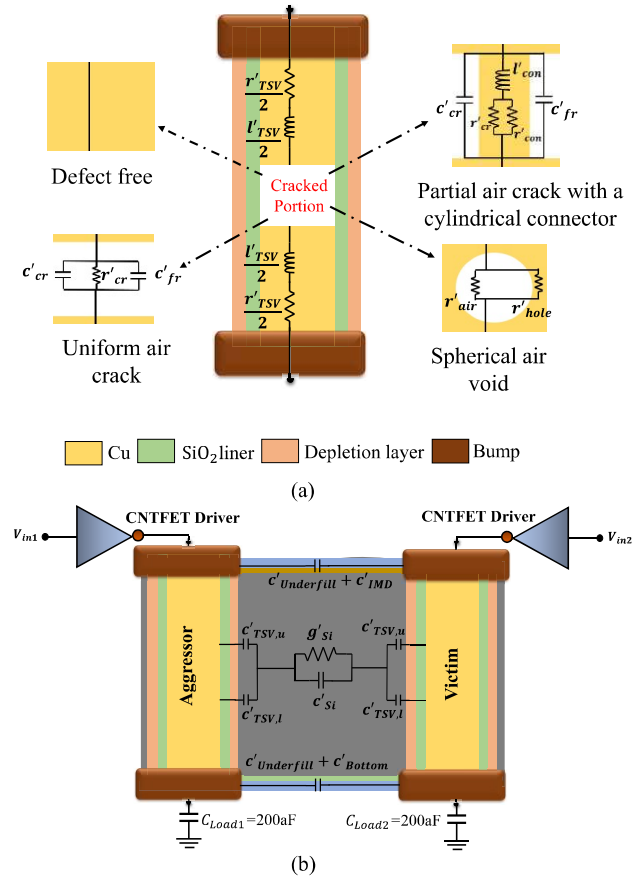


Fig. 1. (a) Equivalent electrical model of a TSV with the zoomed-in view of cracked portion as defect-free, uniform air crack, partial air crack with a cylindrical connector, and spherical air void cases of cylindrical TSVs. (b) Capacitive coupled DVL setup, wherein the aggressor and victim represents the equivalent RLGC model of (a).

the aggressor and victim lines can be represented by Fig. 1(a). As presented in Fig. 1, the TSV filler is surrounded by an oxide liner that separates the Cu filler from the Si substrate. Due to this, the metal–oxide–semiconductor (MOS) effect comes into existence that leads an oxide capacitance to exist. Additionally, a depletion layer comprises of lossless Si is also employed to improve the insulation between the Cu and the Si substrate to reduce the leakage current. Cu is commonly used as a filler material in TSV and bump, that functions as a conductive connector to connect different layers/dies. In addition, an inter metal dielectric (IMD) layer isolates the bump from the Si substrate and an underfill layer made of silica-filled anhydride resin polymers separates the bumps from one another and is used to avoid the interbump leakage. However, the TSV structure is buried in Si substrate that is made of lossy Si material. Table I presents the physical dimensions [7] of a defect-free cylindrical TSV including the scenarios of various types of TSV faults.

Furthermore, the analytical interpretations in terms of RLGC parasitic equations of the above-described models are derived in the following subsections based on the presented VIA configurations of different scenarios (as shown in Fig. 1).

1) *VIA Resistance*: As temperature rises, the increased frequency of collisions between free electrons and the metal ions results in heat dissipation that represents the VIA resistance

TABLE I
TSV PHYSICAL DIMENSIONS AT 7 nm TECHNOLOGY

Parameter	Quantative Value (nm)
Via diameter (d_{TSV})	10.5
Via height (h_{TSV})	30-120
Bump diameter (d_{bump})	22.26
Bump height (h_{bump})	3.71
Oxide liner thickness (t_{ox})	1.05
Depletion layer thickness (t_{dep})	1.65
Via pitch (p_{TSV})	23
IMD height (h_{IMD})	1.05
Crack height (h_{crack})	0.06-1.5
Connector radius (r_{con})	1.05-5.04
Hole radius (r_{hole})	1.05-5.04

(R_{via}). At low frequencies, the R_{via} and L_{via} assume dc values, but as frequency increases, the cross-sectional current distribution in the VIA line becomes nonuniform and moves to the exterior of the conductors due to the skin and proximity effects that become noticeable. However, to model R_{TSV} and R_{Bump} with a nonuniform current distribution, the depth of penetration, i.e., skin depth, has to be determined to calculate the ac resistances of the TSV and bump. Additionally, in parallel VIA lines, the proximity factor (k_p) represents an increase in the apparent $R_{TSV,ac}$ and $R_{Bump,ac}$ beyond the action of the skin effect alone. Therefore, there is an increase in R_{via} due to the existence of $R_{TSV,ac}$ and $R_{Bump,ac}$ that exist due to high-frequency effects called as skin and proximity effects [18]. Therefore, an expression of R_{via} for defect free, uniform air crack, partial air crack, and spherical air void can be obtained from [18] considering $h = h_{TSV}$, $h = (h_{TSV} - h_{crack})$, $h = (h_{TSV} - h_{crack})$, and $h = (h_{TSV} - 2r_{hole})$, respectively.

It is presumed that the air-filled crack occurs in the middle of the uniform air-cracked TSV that results in crack resistance (R_{cr}) [19] and can be derived as

$$R_{cr} = \frac{\rho_{air} \times h_{crack}}{\pi r_{TSV}^2}. \quad (1)$$

However, the expression of R_{cr} of partially air-cracked TSV can be calculated using (1) considering $\rho_{TSV} = \rho_{air}$ and $r_{TSV} = (r_{TSV} - r_{con})$. Also, the connector resistance (R_{con}) can be obtained using (1) considering $h = h_{crack}$ and $r_{TSV} = r_{con}$. In spherical air void defected TSV, due to the presence of an air-filled vacant space (air hole) across the height of $2r_{hole}$, an air resistance (R_{air}) exists that attenuates the transmission of signal. Therefore, R_{air} is obtained using (1) considering $h = 2r_{hole}$, $r_{TSV} = r_{hole}$ and $\rho_{TSV} = \rho_{air}$. However, there also exists a hole resistance (R_{hole}) due to the Cu-filled space that surrounds the spherical air void region and is provided in [15].

2) *VIA Inductance*: In the proposed TSV configuration, inductors are formed due to the concentrated magnetic flux around the loop that are formed by the two parallel conducting TSV lines. Therefore, an equivalent VIA inductance (L_{via}) can be defined as

$$L_{via} = L_{TSV} + L_{Bump} \quad (2)$$

where L_{via} can be obtained from [18] considering $h_{TSV} = (h_{TSV} - h_{crack})$. Similarly, the expression of connector

inductance (L_{con}) of partially air-cracked TSV can be obtained from [18] considering $h_{TSV} = h_{crack}$ and $r_{TSV} = r_{con}$.

3) *Si Conductance*: The conductor loss between the coupled VIA lines is caused by the lossy nature of the conductive Si substrate. Therefore, the Si conductance (G_{Si}) can be modeled considering the coupled VIA lines as two parallel wires. Although, G_{Si} expressions for uniform and partial air-cracked TSVs can be obtained from [18].

However, the expression of G_{Si} in the case of spherical air void can be obtained as

$$G_{Si} = \pi \sigma_{Si} \left[\left\{ \frac{h_{TSV} - 2r_{hole}}{\cosh^{-1} \left(\frac{p_{TSV}^2 - 2r_{TSV}^2}{2r_{TSV}^2} \right)} \right\} + \left\{ \frac{2r_{hole}}{\cosh^{-1} \left(\frac{p_{TSV}^2 - 2(r_{TSV} - r_{hole})^2}{2(r_{TSV} - r_{hole})^2} \right)} \right\} \right]. \quad (3)$$

4) *Capacitance*: There are numerous kinds of capacitive parasitics associated with the cylindrical TSV having defect free and various defected conditions that can be explained as follows.

a) *Substrate capacitance*: Since the Si is a semiconductor with a considerable relative permittivity value that causes the existence of the capacitive coupling between the VIA lines. Therefore, an equivalent substrate capacitance (C_{Si}) can be modeled as parallel wire capacitor [20] and can be obtained from [2] for uniform and partial air-cracked TSVs.

However, considering the defects, C_{Si} for the spherical air void TSV can be expressed as

$$C_{Si} = 2\pi \epsilon_0 \epsilon_{r,si} \left[\left\{ \frac{h_{TSV} - 2r_{hole}}{\cosh^{-1} \left(\frac{p_{TSV}^2 - 2r_{TSV}^2}{2r_{TSV}^2} \right)} \right\} + \left\{ \frac{2r_{hole}}{\cosh^{-1} \left(\frac{p_{TSV}^2 - 2(r_{TSV} - r_{hole})^2}{2(r_{TSV} - r_{hole})^2} \right)} \right\} \right]. \quad (4)$$

b) *Underfill, IMD, and bottom capacitances*: Considering a coupled VIA configuration (as shown in Fig. 1), an underfill capacitance ($C_{underfill}$) can be considered between the bumps due to an underfill layer. Likewise, the presence of IMD and bottom oxide layers between the upper parts and lower parts of the TSVs forms the IMD and bottom capacitances, respectively (C_{IMD} and C_{Bottom}). Therefore, the $C_{underfill}$, C_{IMD} , and C_{Bottom} can be derived using parallel wire capacitance model [20] and hence can be obtained from [2] and [7].

c) *Upper and lower TSV capacitances*: As presented in Fig. 1, the upper (or lower) TSV capacitance is the parallel combination of liner, upper (or lower) bump, and upper (or lower) fringing capacitances. Hence, the upper and lower TSV capacitances ($C_{TSV,u}$ and $C_{TSV,l}$) can be defined as

$$C_{TSV,u} = C_{ins}/2 + C_{Bump1} + C_{fringe(upper)} \quad (5)$$

$$C_{TSV,l} = C_{ins}/2 + C_{Bump2} + C_{fringe(lower)} \quad (6)$$

where the C_{ins} is the series connection of oxide (C_{ox}) and depletion (C_{dep}) capacitances.

The oxide and depletion capacitances (C_{ox} and C_{dep}) are formed between the VIA filler and the Si substrate due to the presence of oxide and depletion layers, respectively. Therefore, C_{ox} and C_{dep} can be obtained using coaxial cable capacitance model [21] and are provided in [7].

However, the expressions of C_{ox} and C_{dep} for spherical air void-defected TSV can be calculated as

$$C_{ox} = 2\pi\epsilon_0\epsilon_{r,ox} \left[\left\{ \frac{h_{TSV} - h_{IMD} - h_{crack}}{\ln\left(\frac{r_{TSV} + t_{ox}}{r_{TSV}}\right)} \right\} + \left\{ \frac{2r_{hole}}{\ln\left(\frac{r_{TSV} + t_{ox}}{(r_{TSV} - r_{hole})}\right)} \right\} \right]. \quad (6b)$$

However, the upper bump capacitance (C_{Bump1}) that is formed

$$C_{dep} = 2\pi\epsilon_0\epsilon_{r,Si} \left[\left\{ \frac{h_{TSV} - h_{IMD} - 2r_{hole}}{\ln\left(\frac{r_{TSV} + t_{ox} + t_{dep}}{(r_{TSV} + t_{ox})}\right)} \right\} + \left\{ \frac{2r_{hole}}{\ln\left(\frac{r_{TSV} + t_{ox} + t_{dep}}{(r_{TSV} - r_{hole} + t_{ox})}\right)} \right\} \right]. \quad (6c)$$

Due to the presence of IMD layer C_{IMD} can be modeled using parallel plate capacitance model and is provided in [7]. Likewise, the lower bump capacitance (C_{Bump2}) between the lower bumps due to a bottom oxide layer can be calculated by replacing $h_{IMD} = t_{ox}$ using the equation provided in [7]. Additionally, the “bump fringing capacitance” occurs between the lateral side of the bump and Si substrate due to the presence of fringing electric field lines across the edges of these conducting regions. The IMD and bottom oxide layers with the combination of an underfill layer, cause the upper and lower bump fringing capacitances [$C_{fringe(upper)}$ and $C_{fringe(lower)}$], respectively. Therefore, $C_{fringe(upper)}$ can be derived as [22]

$$C_{fringe(upper)} = \epsilon_{avg} r_{Bump} \ln \left[\left\{ 16\pi \left(\frac{r_{Bump}}{h_{IMD} + h_{under}} \right) - 1 \right\} \right]. \quad (6d)$$

Similarly, $C_{fringe(lower)}$ can be obtained using (6d) considering $h_{IMD} = t_{ox}$.

Moreover, in partially air-cracked TSV, a connector-Si (C_{con-Si}) capacitance occurs between the VIA filler and the Si substrate due to the presence of air, oxide, and depletion insulating layers in the cracked portion. Hence, C_{con-Si} forms a parallel combination with $C_{TSV,u}$ and $C_{TSV,l}$. Therefore, C_{con-Si} can be modeled using coaxial cable capacitance model [21] and is given as

$$C_{con-Si} = \pi\epsilon_0\epsilon_{r,avg} \frac{h_{crack}}{\ln\left[\frac{r_{TSV} + t_{ox} + t_{dep}}{r_{con}}\right]}. \quad (7)$$

Furthermore, in uniform and partially air-cracked TSVs, the “crack capacitance (C_{cr})” occurs in parallel with a high resistance. Therefore, the C_{cr} in uniform air-cracked TSV can be obtained using parallel plate capacitance model [23] and is defined as

$$C_{cr} = \epsilon_0\epsilon_{r,air} \times \frac{\pi r_{TSV}^2}{h_{crack}} \quad (8)$$

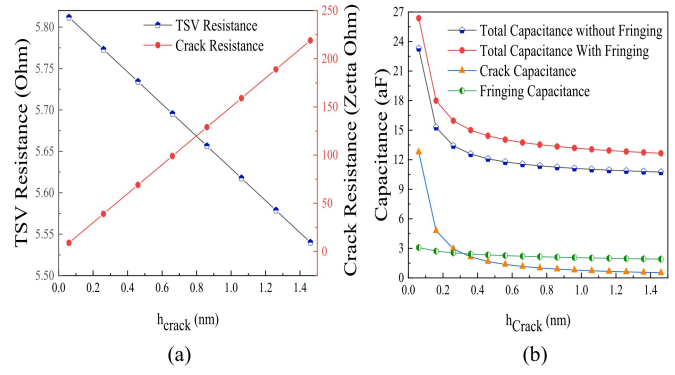


Fig. 2. Variations of (a) R_{TSV} and R_{cr} . (b) Crack, fringing, and total VIA capacitances with and without fringing effects with respect to h_{crack} .

whereas, the C_{cr} for partially air-cracked TSV can be expressed using (8) considering $r_{TSV} = (r_{TSV} - r_{con})$.

Apart from this, due to curvilinear electric field (fringing electric field) close to the lateral sides of the parallel plate structures in the defected TSVs (in the middle of uniform and partial air cracked portions), the fringe capacitance (C_{fr}) occurs due to the presence of air, oxide, and depletion insulating layers. Therefore, C_{fr} for a uniform air-cracked TSV can be obtained using (6d) considering $r_{Bump} = r_{TSV}$ and $(h_{IMD} + h_{under}) = h_{crack}$. However, C_{fr} for a partial air-cracked TSV can be obtained using (6d) by putting $r_{Bump} = (r_{TSV} - r_{con})$.

B. Parasitic Variations

This section presents the variations of RLC parasitics associated with defected portion of the TSV with respect to several defect parameters such as h_{crack} , connecting percentage ($\%k = (r_{con}/r_{TSV}) \times 100$) and void percentage ($\%m = (r_{hole}/r_{TSV}) \times 100$) in the following paragraphs.

1) *Uniform Air Cracked TSV*: It is assumed that a uniform air gap is occurred for the portion of h_{crack} out of entire VIA height in an air crack-defected TSV [as presented in Fig. 1(a)].

Fig. 2(a) presents the variations of R_{TSV} and R_{cr} with respect to h_{crack} . However, Fig. 2(b) represents the variations of C_{cr} , C_{fr} , and $C_{via,total}$ (the parallel combination of C_{cr} and C_{TSV}) with and without fringing effects with the variations in h_{crack} at $h_{TSV} = 30$ nm. It can be depicted that, with an increase in h_{crack} , the R_{cr} increases and R_{TSV} decreases. The primary reason behind this is that, for an increased h_{crack} value, the air-filled cracked region becomes larger compared to the Cu-filled region. However, it can be noticed that the C_{cr} and C_{fr} become smaller with an increased h_{crack} value due a more separation between upper and lower filler regions. Moreover, it can also be noticed that there is a considerable increase in total capacitances when fringing effect is considered. Therefore, fringing capacitance is an important parasitic that should be considered as it affects the overall VIA performance.

2) *Partial Air Cracked TSV With a Cylindrical Connector*: Considering partially air-cracked TSV [as shown in Fig. 1(a)], with a cylindrical connector between the two segments of the Cu-filled regions, it is assumed that the partial air crack occurs in the midsection of VIA height. Fig. 3(a) represents

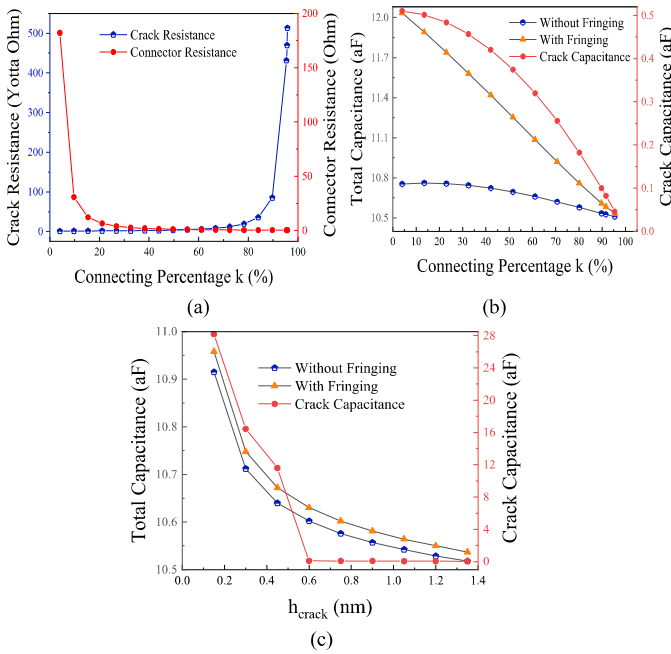


Fig. 3. Variations of (a) R_{cr} and R_{con} . (b) Crack, fringing, and total VIA capacitances with and without fringing effects with respect to connecting percentage ($\%k$). (c) Crack, fringing, and total VIA capacitances with and without fringing effects with respect to h_{crack} .

the variations of crack and connector resistances (R_{cr} and R_{con}) with respect to $\%k$ at $h_{TSV} = 30$ nm. Furthermore, Fig. 3(b) and (c) represents the variations of crack and total capacitance (with and without fringing effect) with respect to $\%k$ and h_{crack} , respectively, at $h_{TSV} = 30$ nm. It is depicted that, for higher $\%k$, the R_{cr} increases due to the reduced surface area of cracked portion while R_{con} reduces due to more surface area of remaining filler region. Moreover, it can be inferred that, all the capacitances are decreasing for an increase in $\%k$ [as shown in Fig. 3(b)] and h_{crack} [as shown in Fig. 3(c)] due to a lesser air-filled surface area and more separation between the upper and lower parts of conducting filler regions, respectively.

3) *Spherical Air Void Defected TSV*: Considering a spherical air void [as depicted in Fig. 1(a)] defected TSV, it is assumed that the spherical air-filled hole is present in the center of the TSV. In this VIA configuration, the air-filled hole is simply modeled by R_{air} whereas the resistance offered by remaining Cu-filled region within the cracked portion is represented by R_{hole} . However, its capacitive parasitic is negligible and hence ignored. Moreover, Fig. 4 represents the variations of R_{hole} and R_{air} with respect to $\%m$ at $h_{TSV} = 30$ nm. It can be observed that the R_{air} decreases while R_{hole} increases for a higher $\%m$. This is due to the fact that, as the $\%m$ varies with increase in r_{hole} , the R_{air} value reduces as the overall surface area of void hole increases. However, the remaining surface area of Cu filled region in faulty part of the TSV becomes less and hence, the R_{hole} increases. Table II presents a list of VIA parasitic for different cases of cylindrical TSV that are obtained based on the different physical parameters of the TSV at 7 nm technology considering the $h_{TSV} = 30$ nm.

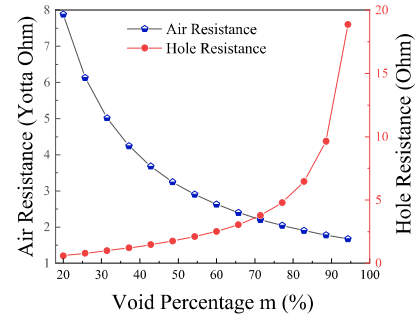


Fig. 4. Variations of hole and air resistances with respect to void percentage ($\%m$).

Additionally, the TSV fault analysis is performed with the structural 3-D full-wave simulation using an Ansoft HFSS tool.

Considering, $h_{TSV} = 30$ nm, $r_{hole} = 5.04$ nm, and $h_{crack} = 1.5$ nm, the electric field (E -field) intensity in the liner layer is analyzed. It can be noticed that the edges of the TSV surface (in defected portions) have more curvature and hence the charge density in that region is larger resulting in C_{fr} . However, the electric field between the parallel plates formed by the conducting regions of TSV metal results in C_{cr} . It can also be noticed that C_{fr} and C_{cr} decrease with an increase in h_{crack} . This is because, the separation increases between the upper and lower conducting regions of TSV metal with an increase in h_{crack} . Also, it was noticed that, since the TSV is cut off due to a uniform crack that results in a very high crack resistance (R_{cr}) and hence, there is complete attenuation of the signal flowing from one terminal to the other. However, there is more distribution of current incase of a spherical void TSV compared to uniform cracked TSV. This is due to the path provided by the remaining conducting region of TSV (in the defected portion) that provides a hole resistance (R_{hole}) which forms a parallel combination with an air resistance (R_{air}), thus, offering a path to the current flow from one terminal to another. It can be inferred that, in case of uniform air crack TSV, with an increase in h_{crack} , the R_{cr} increases offering more R_{via} to the distribution of current in the surface of TSV metal.

Apart from this, in order to validate the RLGC parasitics considering various VIA geometries, the presented TSV models are compared with the existing works [12] and [13]. Considering a uniform air-cracked TSV, the average percentage difference between the values obtained from presented work and [12] for $R_{TSV,dc}$, C_{ox} , and C_{Si} are 9.86%, 34.24%, and 8.24%, respectively. However, in case of a partially air-cracked TSV, the average percentage difference between the values obtained from presented work and [13] for $R_{TSV,dc}$ and C_{ox} are 9.86% and 34.24%, respectively. This is because, in order to achieve a more accurate value of C_{ox} formed by a coaxial cable capacitance model (in both the cases of uniform and partially air crack-defected TSVs) between Cu filled conducting region and Si substrate, the air-filled defected portion (h_{crack}) is eliminated by considering $(h_{TSV} - h_{IMD} - h_{crack})$ in the presented C_{ox} expression. However, in case of a uniform air-cracked TSV, the significant difference in C_{fr} values between

TABLE II
PARASITIC VALUES OF DIFFERENT CASES OF TSV WITH VIA HEIGHT OF 30 nm AT 7 nm TECHNOLOGY

Parasitic	Parasitic Values of cylindrical TSVs with different cases									
	Defect free	Uniform air crack	Partial air crack				Spherical air void			
			$r_{con}=1.05\text{nm}$	$r_{con}=2.1\text{nm}$	$r_{con}=4.2\text{nm}$	$r_{con}=5.04\text{nm}$	$r_{hole}=1.05\text{nm}$	$r_{hole}=2.1\text{nm}$	$r_{hole}=4.2\text{nm}$	$r_{hole}=5.04\text{nm}$
$R_{via}=r'_{via}\times h_{TSV} (\Omega)$	5.9837	5.6925	5.6925	5.6925	5.6925	5.6925	5.5761	5.1684	4.3531	4.0258
$L_{via}=l'_{via}\times h_{TSV} (\text{fH})$	4.7011	4.4795	4.4795	4.4795	4.4795	4.4795	4.3909	4.0806	3.4602	3.2111
$G_{Si}=g'_{Si}\times h_{TSV} (\mu S)$	0.6399	0.6067	0.6067	0.6067	0.6067	0.6067	0.3281	0.3186	0.2814	0.2537
$C_{Si}=c'_{Si}\times h_{TSV} (\text{aF})$	6.7423	6.3930	6.3930	6.3930	6.3930	6.3930	6.9150	6.7145	5.9315	5.346
$C_{ins}=c'_{ins}\times h_{TSV} (\text{aF})$	24.280	23.022	23.022	23.022	23.022	23.022	29.722	21.788	18.101	16.449
$C_{TSV,u}=c'_{TSV,u}\times h_{TSV} (\text{aF})$	21.145	20.516	20.631	20.691	20.882	21.028	20.689	19.899	18.085	17.229
$C_{TSV,l}=c'_{TSV,l}\times h_{TSV} (\text{aF})$	20.961	20.332	20.507	20.507	20.698	20.844	20.505	19.715	17.871	17.045
$R_{cr}=r'_{cr}\times h_{TSV} (\Omega)$	-	2.25×10^{25}	1.41×10^{24}	2.50×10^{24}	2.25×10^{25}	5.63×10^{26}	-	-	-	-
$C_{cr}=c'_{cr}\times h_{TSV} (\text{aF})$	-	0.5108	0.4904	0.4291	0.1839	0.0400	-	-	-	-
$C_{fi}=c'_{fi}\times h_{TSV} (\text{aF})$	-	1.8963	1.0286	0.7261	0.1838	0.0187	-	-	-	-
$C_{con-Si}=c'_{con-Si}\times h_{TSV} (\text{aF})$	-	-	0.2307	0.3508	0.7319	1.0248	-	-	-	-
$R_{con}=r'_{con}\times h_{TSV} (\Omega)$	-	-	7.2793	1.8198	0.455	0.3159	-	-	-	-
$L_{con}=l'_{con}\times h_{TSV} (\text{fH})$	-	-	0.4630	0.3590	0.2550	0.2277	-	-	-	-
$R_{hole}=r'_{hole}\times h_{TSV} (\Omega)$	-	-	-	-	-	-	0.5965	1.334	5.5171	27.061
$R_{air}=r'_{air}\times h_{TSV} (\Omega)$	-	-	-	-	-	-	7.89×10^{24}	3.91×10^{24}	1.97×10^{24}	1.64×10^{24}

the presented work and [12] can be observed. This is due to the fact that, both the oxide and depletion layers are considered as dielectric medium in the presented work.

III. PERFORMANCE COMPARISON

This section delineates a DVL-based simulation setup for the performance analysis of various VIA configurations (as illustrated in Fig. 1). Moreover, the coupled VIA structure is represented by an equivalent RLGC model (as presented in Fig. 1). In this configuration, each VIA line is driven by CNTFET driver with the supply voltage = 0.7 V is used and is terminated with the capacitive loading (C_{Load}) of 200 aF at 7 nm technology. Furthermore, the VIA performance is analyzed for h_{TSV} ranging from 30 to 120 nm. However, h_{crack} , r_{con} , and r_{hole} values are considered as 1.5, 5.04, and 5.04 nm (as presented in Table I), respectively. Additionally, this section analyzes and compares the electrical performance in terms of propagation delay, dynamic power dissipation, peak noise, and crosstalk-induced delay of air gap-defected cylindrical TSVs at 7 nm technology in the following subsections.

A. Impact on Propagation Delay and Power Dissipation

This section examines a comparative analysis of propagation delay and dynamic power dissipation of defect-free and air crack-defected TSV configurations. Based on the switching of the coupled VIA lines, the inputs provided to the TSVs are in-phase switching to investigate the propagation delay

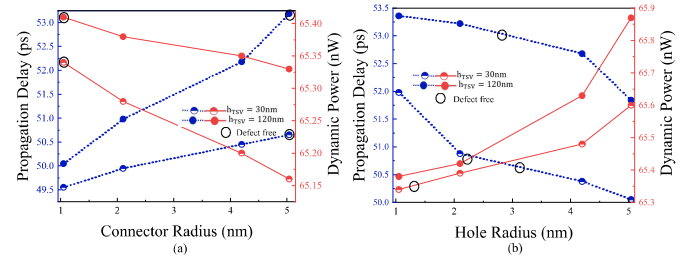


Fig. 5. Propagation delay and dynamic power dissipation of (a) partial air crack and (b) spherical air void defected TSVs.

and dynamic power [25]. Fig. 5(a) and (b) illustrates the propagation delay and dynamic power for partially air crack and spherical air void defected TSV structures, respectively, considering h_{TSV} of 30 and 120 nm. It is observed that, in the case of a partially air-cracked TSV for an increasing r_{con} in the range of 1.05–5.04 nm, the propagation delay of VIA lines meets the condition of defect-free TSV. The primary reason behind approaching toward the optimal result is due to an increased value of connector-to-Si capacitance and crack resistance. It is primarily due to a lesser separation between filler and substrate and a reduced surface area of cracked portion, respectively (as observed from Table II). However, the 0.098% and 0.206% improvements in delay considering $h_{TSV} = 30$ and 120 nm, respectively, can be observed at $r_{con} = 5.04$ nm to approach the defect-free condition. Similarly, considering a spherical air void-defected

TSV, for a variation of r_{hole} ranging from 1.05 to 5.04 nm, the propagation delay reduces until it reaches to the value of defect-free condition, later it is decreased further. This is due to the fact that air resistance is reducing but the hole resistance is increasing that leads to an overall decrease in the VIA resistance. Also, the existence of an air void hole reduces the coupling capacitance between coupled VIA lines resulting in reduced delay. Additionally, it can be inferred that the partially cracked TSV consumes less power with an increase in r_{con} that results in overall reduction of the VIA resistance. The reason behind this is that the surface area of cracked portion lowers while the Cu-filled area of same portion enhances that results in an increased R_{cr} and decreased R_{con} values. Hence, the parallel combination of these resistances results in overall reduction of total VIA resistance. Therefore, approaching to an ideal condition (when $r_{\text{con}} = 5.04$ nm and becomes closed to VIA radius of defect-free TSV) the overall percentage improvements in power dissipation are 0.276% and 0.122% at $h_{\text{TSV}} = 30$ and 120 nm, respectively. However, the spherical air void-defected TSV consumes more power with the increase in an r_{hole} that results in increased value of the hole resistance. However, the spherical air void-defected TSV consumes more power with the increase in an r_{hole} that results in increased value of the hole resistance. This is because, the remaining Cu-filled surface area becomes lesser for defected portion resulting in an increased R_{hole} value. Although, considering a small hole ($r_{\text{con}} = 1.05$ nm), the dynamic power is improved by 0.076% and 0.045% at $h_{\text{TSV}} = 30$ and 120 nm, respectively, but still spherical air void defected TSV takes larger propagation delay as 2.62% and 0.319% when compared to an ideal scenario. Consequently, it is observed from Fig. 5 that the significant improvement in propagation delay and power dissipation for a defect-free TSV at $h_{\text{TSV}} = 30$ nm are 5.01% and 0.10%, respectively, compared to $h_{\text{TSV}} = 120$ nm. Additionally, the effect of propagation delay and power consumption as a percentage change in the power delay product (PDP) of defected TSVs over defect-free TSV is illustrated in Table III. Considering a partially cracked TSV, it is observed that there is significant improvement in PDP as 0.27% and 0.12% at $h_{\text{TSV}} = 30$ and 120 nm, respectively, when it approaches an ideal condition at $r_{\text{con}} = 5.04$ nm. Also, the considerable reduction in PDP can be observed at lower VIA heights. However, the substantial reduction in PDP can be observed as 55.56% at $h_{\text{TSV}} = 30$ nm compared to $h_{\text{TSV}} = 120$ nm. Conversely, the performance degradation can be observed in case of spherical air void-defected TSV when r_{hole} varies from 1.05 to 5.04 nm range. However, it can be inferred that even a small air void with $r_{\text{hole}} = 1.05$ nm causes 2.72% and 2.57% increase in PDP compared to defect-free TSV at $h_{\text{TSV}} = 30$ nm and $h_{\text{TSV}} = 120$ nm, respectively. Considering a uniform air-cracked TSV, the signal is attenuated completely due to the presence of very high crack resistance.

B. Impact on Peak Noise and Crosstalk Delay

This sub-section presents the peak noise and crosstalk-induced delay analysis of cylindrical TSV with consideration

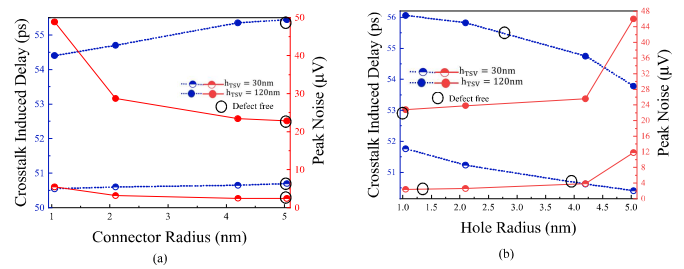


Fig. 6. Peak noise and crosstalk induced delay of (a) partial air crack and (b) spherical air void defected TSVs.

of different kinds of air crack defects by using the DVL setup as depicted in Fig. 1. Based on the applied out-phase switching transitions and coupling between the VIA lines, the line that experiences more delay acts as a victim line while the other as an aggressor [24], [25]. Therefore, the dynamic crosstalk delay is investigated with a considerable Miller coupling factor (MCF) for different VIA cases [26]. However, the functional crosstalk is evaluated when the aggressor line is supplied with a pulse input signal and the victim line is grounded. Therefore, the peak noise can be observed in the form of undesired spikes at the victim line that is accountable for circuitry malfunctions. For partially air-cracked and spherical air void-defected TSVs, the peak noise and crosstalk-induced delay at VIA heights of 30 and 120 nm are shown in Fig. 6(a) and (b), respectively. In case of partial air-cracked TSV [as shown in Fig. 6(a)], the results indicate that with the variations of r_{con} from 1.05 to 5.04 nm, the effect of functional crosstalk on victim line reduces. Therefore, the applied signal propagates through VIA line appropriately with reduced interference between the coupled vias and hence victim line experiences lesser voltage spikes. Therefore, it meets the condition of defect-free TSV with the amount of peak noise close to 2.48 and 22.3 μV at h_{TSV} of 30 and 120 nm, respectively. This is due to the fact that the resistance (R_{con}) offered by the Cu-filled conducting region within the cracked portion becomes lower along with the reduction in the total VIA resistance. Also, the smaller crack and fringing capacitances (C_{cr} and C_{fr}) results in reduction of loss of signal propagation. Therefore, approaching toward the defect-free condition, the considerable improvement in peak noise can be observed as 4.83% and 2.24% at h_{TSV} of 30 and 120 nm, respectively. Also, considering a spherical air void-defected TSV, the increased size of air void (as r_{hole} varies from 1.05 to 5.04 nm) causes more peak noise across a victim line. However, even with a small air void (at $r_{\text{hole}} = 1.05$ nm), the peak noise is increased by 0.24% and 0.82% at h_{TSV} of 30 and 120 nm, respectively, with respect to defect free case. Hence, this defected case approaches to the ideal condition when r_{hole} becomes negligible. Table IV signifies the percentage difference in peak noise of defected TSVs while compared with defect free case. Wherein, the percentage difference in peak noise of partially cracked and spherical air void defected TSVs (with r_{con} and r_{hole} as 2.1 nm) compared to defect-free condition are 24.23% and 1.15%, respectively, at h_{TSV} of 30 nm. Furthermore, it can be noticed that, in partial air-cracked TSV, crosstalk delay increases with increasing

TABLE III
PERCENTAGE CHANGE IN PDP OF DEFECTED TSV OVER DEFECT FREE TSV

h_{TSV} (nm)	% Power Delay Product (PDP)							
	Air Crack with Connector				Spherical Air Void			
	$r_{con}=1.05\text{nm}$	$r_{con}=2.1\text{nm}$	$r_{con}=4.2\text{nm}$	$r_{con}=5.04\text{nm}$	$r_{hole}=1.05\text{nm}$	$r_{hole}=2.1\text{nm}$	$r_{hole}=4.2\text{nm}$	$r_{hole}=5.04\text{nm}$
30	-2.12	-1.21	-0.60	-0.27	2.72	2.41	1.58	0.60
60	-2.11	-1.81	-0.52	-0.24	2.71	2.36	1.39	0.53
90	-2.09	-1.78	-0.44	-0.18	2.69	2.29	1.33	0.44
120	-1.86	-1.03	-0.31	-0.12	2.57	2.17	1.27	0.29

TABLE IV
PERCENTAGE DIFFERENCE IN PEAK NOISE OF DEFECTED TSV COMPARED TO DEFECT FREE CASE

h_{TSV} (in nm)	% Change in Peak Noise of Defected Conditions w.r.t. Defect Free Condition							
	Air Crack with Connector				Spherical Air Void			
	$r_{con}=1.05\text{nm}$	$r_{con}=2.1\text{nm}$	$r_{con}=4.2\text{nm}$	$r_{con}=5.04\text{nm}$	$r_{hole}=1.05\text{nm}$	$r_{hole}=2.1\text{nm}$	$r_{hole}=4.2\text{nm}$	$r_{hole}=5.04\text{nm}$
30	111.9	24.2	4.3	-4.2	0.3	1.1	12.2	106.2
60	109.0	22.1	2.8	-1.3	0.7	2.4	14.8	133.5
90	109.2	22.9	1.9	-0.7	0.8	3.5	22.6	191.7
120	119.2	28.7	1.6	-0.3	0.9	6.7	46.5	353.8

TABLE V
PERCENTAGE DIFFERENCE IN CROSSTALK DELAY OF DEFECTED TSV COMPARED TO DEFECT FREE CASE

h_{TSV} (nm)	% Change in Crosstalk Delay of Defected Conditions w.r.t. Defect Free Condition							
	Air Crack with Connector				Spherical Air Void			
	$r_{con}=1.05\text{nm}$	$r_{con}=2.1\text{nm}$	$r_{con}=4.2\text{nm}$	$r_{con}=5.04\text{nm}$	$r_{hole}=1.05\text{nm}$	$r_{hole}=2.1\text{nm}$	$r_{hole}=4.2\text{nm}$	$r_{hole}=5.04\text{nm}$
30	-1.3	-0.2	-0.14	-0.12	1.0	0.6	-1.6	-3.9
60	-1.7	-1.3	-0.37	-0.16	1.1	0.7	-1.3	-2.9
90	-1.6	-0.7	-0.28	-0.09	1.2	0.8	-0.9	-1.7
120	-1.5	-0.5	-0.16	-0.03	1.3	0.9	-0.2	-0.6

variation of r_{con} [as depicted in Fig. 6(a)]. The reasons behind these are, the parasitics C_{con-Si} and R_{cr} become dominant due to reduced separation between the filler and the substrate materials. Moreover, a lower surface area of cracked portion becomes the major cause behind the significant crosstalk delay in coupled partially cracked TSVs. Conversely, in spherical void-defected TSV, the crosstalk delay reduces with an increase in r_{hole} [as shown in Fig. 6(b)]. It is due to the reduced value of coupling capacitance between spherical void-defected coupled VIA lines that contributes to less crosstalk delay. Furthermore, Table V presents the percentage variations in crosstalk-induced delay of defected VIA cases compared to the defect-free TSV. The improvement in crosstalk delay (as presented in Table V) for a partially cracked TSV compared to a defect-free condition is 0.12% and 0.03% at h_{TSV} of 30 and 120 nm, respectively, considering $r_{con} = 5.04$ nm. However, considering a spherical air void defected TSV, even with a

small air void ($r_{hole} = 1.05$ nm.), the substantial increase in crosstalk delay can be observed as 1.04% and 1.31% at h_{TSV} of 30 and 120 nm, respectively. Consequently, it can be seen [as presented in Fig. 6(a)] that the partial air crack with a cylindrical connector defect approaches the ideal condition of defect-free case when the connector radius r_{con} reaches the value of TSV radius r_{TSV} . Also, from Fig. 6(b), it can be depicted that the spherical air void defected TSV becomes closed to the case of defect-free VIA with the value of hole radius r_{hole} is as small as possible. This is due to the fact that the presence of even a small air void introduces the existence of R_{hole} causing an increase in R_{via} . When the R_{hole} becomes zero, the TSV can be considered as a defect free TSV; whereas, the value of R_{hole} approaching infinity, then the defect can be considered as a full open fault. Hence, the increase in the size of void results in the variation of R_{hole} from few ohms to infinity.

IV. CONCLUSION

The article demonstrated a comparative analysis of the several air gap-defected VIA configurations with respect to a defect-free cylindrical TSV considering different VIA heights. Using a DVL arrangement, a T-type equivalent RLGC network is demonstrated for an accurate VIA dimension at 7 nm technology. The above-mentioned model with CNTFET driver and capacitive coupled VIA lines is used for accurate estimation of the VIA performance in terms of peak noise, crosstalk, propagation delay, dynamic power dissipation, and PDP for different VIA configurations. Based on the analysis and investigation, it can be inferred that the TSVs with partially cracked or spherical air void defects with permissible limits of defect parameters are acceptable. However, the uniform air-cracked TSV is not a suitable VIA configuration due to the complete attenuation of the signal within its cracked region. It is observed that the partial air-cracked TSV with a cylindrical connector meets an ideal condition of defect-free TSV as the connector radius (r_{con}) approaches the value of VIA radius (r_{TSV}). However, the spherical void-defected TSV satisfies the above condition when its hole radius (r_{hole}) becomes as minimum as possible. Although, the fabrication has been evolved for different shapes of TSVs but the manufacturing of complete defect-free TSVs is still a challenging task. Therefore, this research work addressed the feasibilities of some sorts of defected TSVs to provide the performance comparison considering an air gap defect of different cracked shapes for future research in 3-D ICs.

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